Kalman Delta Networks 2:
Two-Axis Factorized Uncertainty

August 28, 2026

Abstract

Kalman Delta Networks treat a recurrent key–value memory as the posterior mean of a latent linear map. Existing variants retain uncertainty along only one axis. Diagonal KDN uses a key-space covariance shared by every value coordinate, $\boldsymbol{I}_{d_v} \otimes \boldsymbol{P}_t$, whereas a complementary value-wise isotropic model uses $\boldsymbol{B}_t \otimes \boldsymbol{I}_{d_k}$. This note first derives the value-wise model and its independent scalar information scans. It then introduces KDN-2, whose retained covariance is the two-axis separable family

$$\boldsymbol{\Sigma}_t = \boldsymbol{B}_t \otimes \boldsymbol{P}_t,$$

with both factors positive diagonal. The exact conditional mean has a simple rank-one form: a key-wise uncertainty direction multiplies value-wise uncertainty gates. The exact posterior covariance, however, is generally not another single Kronecker product. We derive the exact posterior, characterize the exceptional closure condition, and obtain the joint reverse-KL projection back to the separable family. That projection is a positive matrix-scaling problem, unique only up to an intrinsic factor gauge. Consequently, KDN-2 is a principled two-axis uncertainty model, but it does not inherit the independent Möbius scans or the single shared compact-WY memory system of one-axis KDN without an additional approximation. We also connect KDN-2 to Gated DeltaNet-2: both expose key- and value-axis modulation, but KDN-2 ties erasure and writing through one innovation, and their generic algebraic intersection is the scalar KDA rule.

1 Scope and notation

Let $K = d_k$ and $V = d_v$. The recurrent memory $\boldsymbol{S}_t \in \mathbb{R}^{K\times V}$ maps a key to a value. Its jth value column is $\boldsymbol{s}_{t,j} \in \mathbb{R}^K$, and column-major vectorization is

$$\boldsymbol{x}_t = \operatorname{vec}(\widetilde{\boldsymbol{S}}_t) = \begin{bmatrix} \widetilde{\boldsymbol{s}}_{t,1}^{\mathsf{T}} & \cdots & \widetilde{\boldsymbol{s}}_{t,V}^{\mathsf{T}} \end{bmatrix}^{\mathsf{T}}. \tag{1}$$

A tilde denotes the latent random memory, a bare symbol denotes its retained mean, and a hat denotes the exclusive one-step prediction before the current observation is assimilated. The observation model is

$$\boldsymbol{v}_t = \widetilde{\boldsymbol{S}}_t^{\mathsf{T}} \boldsymbol{k}_t + \boldsymbol{e}_t = \boldsymbol{H}_t \boldsymbol{x}_t + \boldsymbol{e}_t, \qquad \boldsymbol{H}_t = \boldsymbol{I}_V \otimes \boldsymbol{k}_t^{\mathsf{T}}, \qquad \boldsymbol{e}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_t). \tag{2}$$

Unless stated otherwise, $\boldsymbol{R}_t = \operatorname{diag}(r_{t,1}, \ldots, r_{t,V}) \succ 0$. Diagonal observation noise preserves cross-value independence while allowing different value coordinates to have different reliability.

The two one-axis uncertainty families are

$$\underbrace{\boldsymbol{I}_V \otimes \boldsymbol{P}_t}_{\text{shared key-space uncertainty}} \qquad \text{and} \qquad \underbrace{\boldsymbol{B}_t \otimes \boldsymbol{I}_K}_{\text{value-wise isotropic uncertainty}}, \tag{3}$$

where $\boldsymbol{P}_t \in \mathbb{R}^{K\times K}$ and $\boldsymbol{B}_t = \operatorname{diag}(b_{t,1}, \ldots, b_{t,V})$. The first is the family used by Diagonal KDN; the second is derived next.

2 Value-wise isotropic uncertainty

2.1 Retained family and state-space model

The value-wise isotropic family assigns one scalar key-space variance to every value coordinate:

$$\widetilde{\boldsymbol{s}}_{t,j} \mid \mathcal{F}_t \sim \mathcal{N}(\boldsymbol{s}_{t,j}, b_{t,j}\boldsymbol{I}_K), \qquad \operatorname{Cov}\Big(\operatorname{vec}(\widetilde{\boldsymbol{S}}_t) \mid \mathcal{F}_t\Big) = \boldsymbol{B}_t \otimes \boldsymbol{I}_K. \tag{4}$$

The columns remain conditionally independent, but their uncertainty need not be equal. Consider the columnwise process and observation model

$$\begin{aligned} \widetilde{\boldsymbol{s}}_{t,j} &= \boldsymbol{D}_t \widetilde{\boldsymbol{s}}_{t-1,j} + \boldsymbol{w}_{t,j}, \quad && \boldsymbol{w}_{t,j} \sim \mathcal{N}(\boldsymbol{0}, \omega_{t,j}\boldsymbol{I}_K), \\ v_{t,j} &= \boldsymbol{k}_t^{\top} \widetilde{\boldsymbol{s}}_{t,j} + e_{t,j}, && e_{t,j} \sim \mathcal{N}(0, r_{t,j}), \qquad \boldsymbol{D}_t = \operatorname{diag}(\boldsymbol{\alpha}_t). \end{aligned} \tag{5}$$

The noises are independent across value coordinates and time. The current shared-noise model is recovered by tying $\omega_{t,j} = \omega_t$ and $r_{t,j} = r_t$.

The mean prediction is

$$\widehat{\boldsymbol{S}}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1}. \tag{6}$$

The channel-wise transition generally makes the exact predictive covariance anisotropic even when the retained covariance is isotropic. Define its average squared transition

$$a_t = \frac{1}{K}\operatorname{Tr}(\boldsymbol{D}_t \boldsymbol{D}_t^{\top}) = \frac{1}{K}\sum_{i=1}^{K} \alpha_{t,i}^2. \tag{7}$$

Projecting the predictive covariance to its average marginal variance gives

$$\widehat{b}_{t,j} = a_t b_{t-1,j} + \omega_{t,j}, \qquad \widehat{\boldsymbol{P}}_{t,j} = \widehat{b}_{t,j}\boldsymbol{I}_K. \tag{8}$$

This prediction is exact when $\boldsymbol{D}_t\boldsymbol{D}_t^{\top} = a_t\boldsymbol{I}_K$ and is otherwise the covariance-coordinate trace projection used by Isotropic KDN.

2.2 Exact conditional mean

Let

$$\delta_{t,j} = v_{t,j} - \boldsymbol{k}_t^{\top}\widehat{\boldsymbol{s}}_{t,j}. \tag{9}$$

The predictive cross-covariance and innovation variance are

$$\operatorname{Cov}(\widetilde{\boldsymbol{s}}_{t,j}, v_{t,j} \mid \mathcal{F}_{t-1}) = \widehat{b}_{t,j}\boldsymbol{k}_t, \qquad \Delta_{t,j} = r_{t,j} + \widehat{b}_{t,j}\|\boldsymbol{k}_t\|_2^2. \tag{10}$$

Therefore column j has scalar gain

$$\beta_{t,j} = \frac{\widehat{b}_{t,j}}{\Delta_{t,j}}, \qquad \boldsymbol{s}_{t,j} = \widehat{\boldsymbol{s}}_{t,j} + \beta_{t,j}\boldsymbol{k}_t\delta_{t,j}. \tag{11}$$

Every value column writes along the same key direction, but each column uses a different uncertainty-derived contraction. In matrix form, with $\boldsymbol{\beta}_t = (\beta_{t,1}, \ldots, \beta_{t,V})^{\top}$,

$$\boldsymbol{S}_t = \widehat{\boldsymbol{S}}_t + \boldsymbol{k}_t\left[\boldsymbol{\beta}_t \odot \left(\boldsymbol{v}_t - \widehat{\boldsymbol{S}}_t^{\top}\boldsymbol{k}_t\right)\right]^{\top}. \tag{12}$$

Equivalently, grouping the predicted memory columnwise gives

$$\boldsymbol{s}_{t,j} = \left(\boldsymbol{I}_K - \beta_{t,j}\boldsymbol{k}_t\boldsymbol{k}_t^{\top}\right)\widehat{\boldsymbol{s}}_{t,j} + \beta_{t,j}\boldsymbol{k}_t v_{t,j}, \qquad j = 1, \ldots, V. \tag{13}$$

2.3 Reverse-KL projection and information scan

The exact posterior precision for column j is

$$\boldsymbol{C}^{\star}_{t,j} = \widehat{c}_{t,j}\boldsymbol{I}_K + \frac{1}{r_{t,j}}\boldsymbol{k}_t\boldsymbol{k}_t^{\mathsf{T}}, \qquad \widehat{c}_{t,j} = \widehat{b}_{t,j}^{-1}. \tag{14}$$

It is dense for a dense key, so it must be projected before the next token.

Proposition 1 (Value-wise isotropic reverse-KL projection)**.** *Let $\pi_{t,j}$ be the exact Gaussian posterior defined by Equations (5)–(14). Over*

$$q_{t,j} = \mathcal{N}(\boldsymbol{m}_{t,j}, b_{t,j}\boldsymbol{I}_K),$$

the unique minimizer of $\mathrm{KL}(q_{t,j} \,\|\, \pi_{t,j})$ *is*

$$\boldsymbol{m}_{t,j} = \boldsymbol{s}_{t,j}, \qquad b_{t,j} = \frac{K}{\mathrm{Tr}(\boldsymbol{C}^{\star}_{t,j})} = \left(\widehat{c}_{t,j} + \frac{\|\boldsymbol{k}_t\|_2^2}{K r_{t,j}}\right)^{-1}. \tag{15}$$

Proof. For a Gaussian target with precision $\boldsymbol{C}^{\star}_{t,j}$, the covariance-dependent part of the reverse KL is $\frac{1}{2}\{b_{t,j}\,\mathrm{Tr}(\boldsymbol{C}^{\star}_{t,j}) - K\log b_{t,j}\}$. Its derivative vanishes only at $b_{t,j} = K/\,\mathrm{Tr}(\boldsymbol{C}^{\star}_{t,j})$. The mean-dependent quadratic is uniquely minimized by the exact posterior mean in Equation (11). □

Equation (15) is the literal variational update. As in Isotropic KDN, an information scale $\mu > 0$ can instead be introduced only in the retained post-write precision:

$$\widehat{c}_{t,j} = \frac{c_{t-1,j}}{a_t + \omega_{t,j}c_{t-1,j}}, \qquad u_{t,j} = \mu\frac{\|\boldsymbol{k}_t\|_2^2}{K r_{t,j}}, \qquad c_{t,j} = \widehat{c}_{t,j} + u_{t,j}. \tag{16}$$

Here $\mu = 1$ is the reverse-KL optimum. Values $\mu \neq 1$ calibrate future protection while leaving the current gain in Equation (11) unchanged.

Substitution gives a scalar Möbius recurrence for every value coordinate:

$$c_{t,j} = \frac{(1 + u_{t,j}\omega_{t,j})c_{t-1,j} + u_{t,j}a_t}{\omega_{t,j}c_{t-1,j} + a_t}, \qquad \boldsymbol{M}_{t,j} = \begin{bmatrix} 1 + u_{t,j}\omega_{t,j} & u_{t,j}a_t \\ \omega_{t,j} & a_t \end{bmatrix}. \tag{17}$$

Thus the uncertainty stage is V independent associative scalar scans.

2.4 When the model is genuinely different

If $b_{0,j}$, $\omega_{t,j}$, and $r_{t,j}$ are tied across j, then induction through Equations (8) and (16) keeps every $b_{t,j}$ and $\beta_{t,j}$ equal. Realized values cannot break this symmetry because Gaussian covariance updates do not depend on the realized observation. At least one value-wise initial variance, process uncertainty, or observation noise must vary for the model to differ from Isotropic KDN.

The sequential mean update still costs $O(KV)$, but the transition applied to column j is

$$\boldsymbol{A}_{t,j} = (\boldsymbol{I}_K - \beta_{t,j}\boldsymbol{k}_t\boldsymbol{k}_t^{\mathsf{T}})\boldsymbol{D}_t. \tag{18}$$

Because $\boldsymbol{A}_{t,j}$ varies with j, the single compact-WY system shared by all value columns in current KDN kernels no longer applies. An exact parallel implementation needs value-batched systems or a grouped-value approximation.

3 KDN-2: combined key–value factorized uncertainty

3.1 Separable diagonal family

KDN-2 combines the two one-axis families by retaining

$$
\begin{aligned}
\boldsymbol{P}_t &= \operatorname{diag}(p_{t,1}, \ldots, p_{t,K}) \succ 0, & \boldsymbol{B}_t &= \operatorname{diag}(b_{t,1}, \ldots, b_{t,V}) \succ 0, \\
\boldsymbol{\Sigma}_t &= \boldsymbol{B}_t \otimes \boldsymbol{P}_t = (\boldsymbol{B}_t \otimes \boldsymbol{I}_K)(\boldsymbol{I}_V \otimes \boldsymbol{P}_t), & \boldsymbol{J}_t &= \boldsymbol{\Sigma}_t^{-1} = \boldsymbol{G}_t \otimes \boldsymbol{C}_t,
\end{aligned}
\tag{19}
$$

where $\boldsymbol{G}_t = \boldsymbol{B}_t^{-1}$ and $\boldsymbol{C}_t = \boldsymbol{P}_t^{-1}$. Hence the covariance and information forms are the same family: inversion simply inverts both diagonal factors. Entry (i, j) of the latent memory has variance

$$
\operatorname{Var}(\widetilde{S}_{t,ij} \mid \mathcal{F}_t) = p_{t,i} b_{t,j}. \tag{20}
$$

The model has $K + V - 1$ covariance degrees of freedom, not KV, because

$$
\boldsymbol{B}_t \otimes \boldsymbol{P}_t = (\lambda \boldsymbol{B}_t) \otimes (\boldsymbol{P}_t / \lambda) \qquad \text{for every } \lambda > 0. \tag{21}
$$

We fix this internal factor gauge by imposing

$$
\frac{1}{V} \sum_{j=1}^{V} \log b_{t,j} = 0 \qquad \Longleftrightarrow \qquad \det(\boldsymbol{B}_t) = 1, \tag{22}
$$

and compensating $\boldsymbol{P}_t$ so that $\boldsymbol{B}_t \otimes \boldsymbol{P}_t$ is unchanged.

At the covariance-family level, the nested special cases are

$$
\begin{array}{rcl}
\boldsymbol{B}_t = \bar{b}_t \boldsymbol{I}_V, \ \boldsymbol{R}_t = r_t \boldsymbol{I}_V & \Longrightarrow & \text{Diagonal KDN}, \\
\boldsymbol{P}_t = \bar{p}_t \boldsymbol{I}_K & \Longrightarrow & \text{value-wise isotropic KDN}, \\
\boldsymbol{B}_t = \bar{b}_t \boldsymbol{I}_V, \ \boldsymbol{P}_t = \bar{p}_t \boldsymbol{I}_K, \ \boldsymbol{R}_t = r_t \boldsymbol{I}_V & \Longrightarrow & \text{Isotropic KDN}.
\end{array}
\tag{23}
$$

The scalar factor in each row can be absorbed into the other covariance factor before applying the gauge in Equation (22). These are one-step covariance-family limits. Recovering the complete one-axis recurrences additionally requires constraining the corresponding factor and using its one-axis predictive projection; in particular, the channel-wise transition in Equation (25) does not preserve $\boldsymbol{P}_t \propto \boldsymbol{I}_K$ without the trace projection in Equation (8).

3.2 Factor-preserving prediction

The key-space mean transition remains

$$
\widehat{\boldsymbol{S}}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1}, \qquad \boldsymbol{D}_t = \operatorname{diag}(\boldsymbol{\alpha}_t). \tag{24}
$$

A simple retained prediction updates the two covariance factors by

$$
\widehat{\boldsymbol{P}}_t = \boldsymbol{D}_t \boldsymbol{P}_{t-1} \boldsymbol{D}_t^{\mathsf{T}} + \boldsymbol{\Omega}_t^{(k)}, \qquad \widehat{\boldsymbol{B}}_t = \boldsymbol{B}_{t-1} + \boldsymbol{\Omega}_t^{(v)}, \tag{25}
$$

with positive diagonal process factors. This equation should be read as a factor-preserving predictive parameterization. It corresponds to the positive-semidefinite full process covariance

$$
\boldsymbol{Q}_t = \boldsymbol{B}_{t-1} \otimes \boldsymbol{\Omega}_t^{(k)} + \boldsymbol{\Omega}_t^{(v)} \otimes (\boldsymbol{D}_t \boldsymbol{P}_{t-1} \boldsymbol{D}_t^{\mathsf{T}}) + \boldsymbol{\Omega}_t^{(v)} \otimes \boldsymbol{\Omega}_t^{(k)}, \tag{26}
$$

because adding Equation (26) to the transitioned prior gives $\widehat{\boldsymbol{B}}_t \otimes \widehat{\boldsymbol{P}}_t$. A generic additive separable process covariance would instead produce a sum of Kronecker products and leave the retained family.

After prediction, apply the gauge normalization in Equation (22). This normalization leaves the full predictive covariance and the full Kalman gain unchanged. It does rescale the factors used below: under $(\widehat{\boldsymbol{B}}_t, \widehat{\boldsymbol{P}}_t) \mapsto (\lambda \widehat{\boldsymbol{B}}_t, \widehat{\boldsymbol{P}}_t / \lambda)$, $\boldsymbol{g}_t \mapsto \boldsymbol{g}_t / \lambda$ and $\boldsymbol{\beta}_t \mapsto \lambda \boldsymbol{\beta}_t$, so their product is invariant.

3.3 Exact innovation and conditional mean

Define

$$\boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t, \qquad z_t = \boldsymbol{k}_t^\top \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t, \qquad \boldsymbol{\delta}_t = \boldsymbol{v}_t - \widehat{\boldsymbol{S}}_t^\top \boldsymbol{k}_t. \tag{27}$$

The innovation covariance and state–observation cross-covariance are

$$\boldsymbol{\Delta}_t = z_t \widehat{\boldsymbol{B}}_t + \boldsymbol{R}_t, \qquad \operatorname{Cov}(\boldsymbol{x}_t, \boldsymbol{v}_t \mid \mathcal{F}_{t-1}) = \widehat{\boldsymbol{B}}_t \otimes \boldsymbol{g}_t. \tag{28}$$

The exact vectorized Kalman gain is therefore

$$\boldsymbol{K}_t = (\widehat{\boldsymbol{B}}_t \otimes \boldsymbol{g}_t)\boldsymbol{\Delta}_t^{-1}. \tag{29}$$

When $\boldsymbol{R}_t = \operatorname{diag}(r_{t,1}, \dots, r_{t,V})$, define

$$\beta_{t,j} = \frac{\widehat{b}_{t,j}}{r_{t,j} + \widehat{b}_{t,j} z_t}, \qquad \boldsymbol{\beta}_t = (\beta_{t,1}, \dots, \beta_{t,V})^\top. \tag{30}$$

The exact posterior mean has the rank-one matrix update

$$\boldsymbol{S}_t^\star = \widehat{\boldsymbol{S}}_t + \boldsymbol{g}_t(\boldsymbol{\beta}_t \odot \boldsymbol{\delta}_t)^\top. \tag{31}$$

Equivalently, grouping the predicted memory columnwise gives

$$\begin{aligned} \boldsymbol{s}_{t,j}^\star &= \left(\boldsymbol{I}_K - \beta_{t,j} \boldsymbol{g}_t \boldsymbol{k}_t^\top\right) \widehat{\boldsymbol{s}}_{t,j} + \beta_{t,j} \boldsymbol{g}_t v_{t,j} \\ &= \left(\boldsymbol{I}_K - \beta_{t,j} \boldsymbol{g}_t \boldsymbol{k}_t^\top\right) \boldsymbol{D}_t \boldsymbol{s}_{t-1,j} + \beta_{t,j} \boldsymbol{g}_t v_{t,j}, \qquad j = 1, \dots, V. \end{aligned} \tag{32}$$

This equation is the central behavioral generalization. The key factor $\widehat{\boldsymbol{P}}_t$ rotates and rescales the write direction through $\boldsymbol{g}_t$, while the value factor $\widehat{\boldsymbol{B}}_t$ produces relative write strengths across value coordinates. Their absolute scales are gauge-dependent; only $\boldsymbol{g}_t \beta_{t,j}$ and ratios among the $\beta_{t,j}$ are identifiable. Setting $\widehat{\boldsymbol{B}}_t = \boldsymbol{I}_V$ and $\boldsymbol{R}_t = r_t \boldsymbol{I}_V$ recovers the Diagonal-KDN gain; setting $\widehat{\boldsymbol{P}}_t = \boldsymbol{I}_K$ recovers Equation (12).

The one-step predictive negative log likelihood, up to the additive Gaussian constant, is

$$\mathcal{L}_{\text{pred},t} = \frac{1}{2} \sum_{j=1}^{V} \left[\frac{\delta_{t,j}^2}{r_{t,j} + \widehat{b}_{t,j} z_t} + \log(r_{t,j} + \widehat{b}_{t,j} z_t) \right]. \tag{33}$$

Both the residual and its variance use the exclusive pre-write state.

3.4 Exact posterior and failure of Kronecker closure

The exact posterior covariance and precision are

$$\begin{aligned} \boldsymbol{\Sigma}_t^\star &= \widehat{\boldsymbol{B}}_t \otimes \widehat{\boldsymbol{P}}_t - (\widehat{\boldsymbol{B}}_t \boldsymbol{\Delta}_t^{-1} \widehat{\boldsymbol{B}}_t) \otimes (\boldsymbol{g}_t \boldsymbol{g}_t^\top), \\ \boldsymbol{J}_t^\star &= \widehat{\boldsymbol{B}}_t^{-1} \otimes \widehat{\boldsymbol{P}}_t^{-1} + \boldsymbol{R}_t^{-1} \otimes \boldsymbol{k}_t \boldsymbol{k}_t^\top. \end{aligned} \tag{34}$$

For diagonal $\boldsymbol{R}_t$, column j has covariance block

$$\boldsymbol{\Sigma}_{t,j}^\star = \widehat{b}_{t,j} \left(\widehat{\boldsymbol{P}}_t - \beta_{t,j} \boldsymbol{g}_t \boldsymbol{g}_t^\top \right). \tag{35}$$

Different $\beta_{t,j}$ values therefore produce differently shaped key-space covariance blocks, not merely differently scaled copies of one common matrix.

Proposition 2 (Exact separable closure)**.** *Assume $K > 1$, $\widehat{\boldsymbol{P}}_t \succ 0$, and $\boldsymbol{g}_t\boldsymbol{g}_t^\top \neq \boldsymbol{0}$. The blocks in Equation (35) can be written as $b^+_{t,j}\boldsymbol{P}^+_t$ with one common, possibly dense, matrix $\boldsymbol{P}^+_t$ if and only if all $\beta_{t,j}$ are equal. For diagonal observation noise, this is equivalent to*

$$\frac{r_{t,j}}{\widehat{b}_{t,j}} = \rho_t \quad \textit{for every } j, \qquad \textit{or equivalently} \qquad \boldsymbol{R}_t = \rho_t\widehat{\boldsymbol{B}}_t. \tag{36}$$

Proof. If $\beta_{t,j} = \beta_t$, then $\boldsymbol{\Sigma}^\star_{t,j} = \widehat{b}_{t,j}(\widehat{\boldsymbol{P}}_t - \beta_t\boldsymbol{g}_t\boldsymbol{g}_t^\top)$, which is separable. Conversely, suppose two blocks with coefficients $\beta_a \neq \beta_b$ are proportional. Then for some $c > 0$,

$$\widehat{\boldsymbol{P}}_t - \beta_a\boldsymbol{g}_t\boldsymbol{g}_t^\top = c(\widehat{\boldsymbol{P}}_t - \beta_b\boldsymbol{g}_t\boldsymbol{g}_t^\top).$$

If $c \neq 1$, the left rearrangement contains the full-rank matrix $(1-c)\widehat{\boldsymbol{P}}_t$ while the other side has rank at most one, a contradiction. Thus $c = 1$ and $\beta_a = \beta_b$. Finally, $\beta_{t,j} = 1/(r_{t,j}/\widehat{b}_{t,j} + z_t)$, so equality of the gains is equivalent to Equation (36). □

The compatible choice $\boldsymbol{R}_t = \rho_t\widehat{\boldsymbol{B}}_t$ is mathematically closed in the value factor, but it makes that factor cancel from the gain:

$$\boldsymbol{S}^\star_t = \widehat{\boldsymbol{S}}_t + \frac{\boldsymbol{g}_t\boldsymbol{\delta}_t^\top}{\rho_t + z_t}. \tag{37}$$

The value uncertainty then has no effect on the posterior mean. Moreover, the common key-space posterior remains dense for a dense key, so the diagonal $\boldsymbol{P}_t$ restriction still requires the usual KDN projection. Useful value-dependent gains and exact Kronecker closure therefore cannot generally be obtained simultaneously.

3.5 Joint reverse-KL projection

We retain the exact mean in Equation (31) and project only its covariance. Let

$$q_t = \mathcal{N}(\mathrm{vec}(\boldsymbol{S}^\star_t), \boldsymbol{B}_t \otimes \boldsymbol{P}_t), \qquad \boldsymbol{B}_t = \mathrm{diag}(\boldsymbol{b}_t), \quad \boldsymbol{P}_t = \mathrm{diag}(\boldsymbol{p}_t). \tag{38}$$

Define four candidate-factor summaries

$$\begin{aligned} A_B &= \sum_{j=1}^{V}\frac{b_{t,j}}{\widehat{b}_{t,j}}, \quad T_B = \sum_{j=1}^{V}\frac{b_{t,j}}{r_{t,j}}, \\ A_P &= \sum_{i=1}^{K}\frac{p_{t,i}}{\widehat{p}_{t,i}}, \quad Z_P = \sum_{i=1}^{K}p_{t,i}k_{t,i}^2. \end{aligned} \tag{39}$$

Proposition 3 (Joint reverse-KL projection)**.** *Up to the factor gauge in Equation (21), the stationary point of $\mathrm{KL}(q_t \,\|\, \pi^\star_t)$, where $\pi^\star_t$ has covariance Equation (34), satisfies*

$$b_{t,j} = \frac{K}{A_P/\widehat{b}_{t,j} + Z_P/r_{t,j}}, \qquad p_{t,i} = \frac{V}{A_B/\widehat{p}_{t,i} + T_B k_{t,i}^2}. \tag{40}$$

Equivalently, with

$$L_{t,ij} = \frac{1}{\widehat{p}_{t,i}\widehat{b}_{t,j}} + \frac{k_{t,i}^2}{r_{t,j}}, \qquad W_{t,ij} = p_{t,i}b_{t,j}L_{t,ij}, \tag{41}$$

the positive matrix $\boldsymbol{W}_t$ has row sums V and column sums K.

Proof. Using the precision in Equation (34), the covariance-dependent part of twice the reverse KL is

$$A_B A_P + T_B Z_P - K \log\det \boldsymbol{B}_t - V \log\det \boldsymbol{P}_t + \text{const.} \tag{42}$$

Differentiating with respect to $b_{t,j}$ and $p_{t,i}$ yields Equation (40). Equivalently, $\partial/\partial p_{t,i} = 0$ gives $\sum_j W_{t,ij} = V$, and $\partial/\partial b_{t,j} = 0$ gives $\sum_i W_{t,ij} = K$. Multiplying every $b_{t,j}$ by λ and dividing every $p_{t,i}$ by λ leaves both the objective and the full covariance unchanged, which is exactly the factor gauge. □

In log-factor coordinates, Equation (42) is convex and is strictly convex after the gauge in Equation (22) is fixed. The positive fixed point is therefore unique in that gauge. It is a matrix-scaling problem, and alternating updates

$$\begin{aligned} b_{t,j} &\leftarrow \frac{K}{A_P/\widehat{b}_{t,j} + Z_P/r_{t,j}}, \\ p_{t,i} &\leftarrow \frac{V}{A_B/\widehat{p}_{t,i} + T_B k_{t,i}^2}, \end{aligned} \tag{43}$$

followed by gauge normalization solve it without constructing a $K \times V$ matrix. These are the standard alternating matrix-scaling updates and converge to the unique gauge-equivalence class because $L_{t,ij} > 0$. Each sweep costs $O(K+V)$ because $L_{t,ij}$ is the sum of two rank-one terms, one strictly positive and one nonnegative. The projection is nevertheless coupled and iterative at every token. It is not two independent Möbius scans over time.

The one-sided boundaries recover the previous models. Holding $\boldsymbol{B}_t = \widehat{\boldsymbol{B}}_t$ fixed gives

$$p_{t,i} = \left[\widehat{p}_{t,i}^{-1} + \left(\frac{1}{V}\sum_{j=1}^{V} \frac{\widehat{b}_{t,j}}{r_{t,j}}\right) k_{t,i}^2\right]^{-1}. \tag{44}$$

For $\widehat{\boldsymbol{B}}_t = \boldsymbol{I}_V$ and $\boldsymbol{R}_t = r_t \boldsymbol{I}_V$, this is the ordinary Diagonal-KDN reverse-KL update. Holding $\boldsymbol{P}_t = \widehat{\boldsymbol{P}}_t$ fixed gives

$$b_{t,j} = \left[\widehat{b}_{t,j}^{-1} + \frac{z_t}{K r_{t,j}}\right]^{-1}, \tag{45}$$

which is the value-wise isotropic update when $\widehat{\boldsymbol{P}}_t = \boldsymbol{I}_K$. These equations describe one posterior projection. Recovering the full value-wise isotropic recurrence also requires the trace-projected prediction in Equation (8). Applying both one-sided full-information updates independently would double-count the same observation; the coupled projection is what allocates the information between factors.

An optional protection scale may multiply the measurement term $k_{t,i}^2/r_{t,j}$ in Equation (41). If the same scale also tempers the likelihood and changes the current gain, it defines a different Gaussian observation model. If the exact gain in Equation (31) is kept fixed, the scaled projection is instead a retained future-protection calibration, as in the one-axis KDN variants; it is not the reverse-KL optimum for Equation (2).

3.6 A practical reference recurrence

A sequential KDN-2 reference implementation can use the following token step:

1. Predict $\widehat{\boldsymbol{S}}_t$, $\widehat{\boldsymbol{P}}_t$, and $\widehat{\boldsymbol{B}}_t$ with Equations (24) and (25).

2. Fix the internal factor gauge by Equation (22).
3. Compute $\boldsymbol{g}_t$, z_t, $\boldsymbol{\beta}_t$, and the exact posterior mean $\boldsymbol{S}_t^\star$ from Equations (27)–(31).
4. Initialize the covariance projection at the predictive factors and run a fixed number of alternating updates in Equation (43).
5. Reapply the gauge normalization and carry the projected factors to token $t+1$.

With N_{proj} alternating sweeps, uncertainty work is $O(N_{\text{proj}}(K+V))$ per token and the memory update is $O(KV)$. This is suitable as a correctness oracle and as a small-scale experiment. It is not yet a drop-in replacement for the current scan-parallel KDN kernel.

4 Connection to Gated DeltaNet-2

4.1 The two updates side by side

Use the same predicted memory $\widehat{\boldsymbol{S}}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1}$. Gated DeltaNet-2 (GDN-2) can be written as

$$\begin{aligned} \boldsymbol{S}_t^{\text{GDN2}} &= \widehat{\boldsymbol{S}}_t - \boldsymbol{k}_t(\boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t)^\top \widehat{\boldsymbol{S}}_t + \boldsymbol{k}_t(\boldsymbol{w}_t \odot \boldsymbol{v}_t)^\top, \\ \boldsymbol{b}_t^{\text{erase}} &\in \mathbb{R}^K, \qquad \boldsymbol{w}_t \in \mathbb{R}^V, \end{aligned} \tag{46}$$

where $\boldsymbol{b}_t^{\text{erase}}$ is a key-channel erase gate and $\boldsymbol{w}_t$ is a value-channel write gate. Both are predicted directly from the current token and are sigmoid-valued in the default implementation; an optional GDN-2 variant expands only the erase-gate range. The keys are ℓ_2-normalized in both model families. Defining the pseudo-key and pseudo-value

$$\widetilde{\boldsymbol{k}}_t = \boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t, \qquad \widetilde{\boldsymbol{v}}_t = \boldsymbol{w}_t \odot \boldsymbol{v}_t, \tag{47}$$

gives the residual-like form

$$\boldsymbol{S}_t^{\text{GDN2}} = \widehat{\boldsymbol{S}}_t + \boldsymbol{k}_t \left(\widetilde{\boldsymbol{v}}_t - \widehat{\boldsymbol{S}}_t^\top \widetilde{\boldsymbol{k}}_t \right)^\top. \tag{48}$$

This resembles a Kalman residual update algebraically, but its erased read uses $\widetilde{\boldsymbol{k}}_t$ while its injected target is $\widetilde{\boldsymbol{v}}_t$. In general it is not the posterior mean for the original observation $\boldsymbol{v}_t = \widetilde{\boldsymbol{S}}_t^\top \boldsymbol{k}_t + \boldsymbol{e}_t^{\text{obs}}$.

Expanding the KDN-2 mean update in Equation (31) gives

$$\boldsymbol{S}_t^{\text{KDN2}} = \widehat{\boldsymbol{S}}_t - \boldsymbol{g}_t \boldsymbol{k}_t^\top \widehat{\boldsymbol{S}}_t \operatorname{diag}(\boldsymbol{\beta}_t) + \boldsymbol{g}_t(\boldsymbol{\beta}_t \odot \boldsymbol{v}_t)^\top, \qquad \boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t. \tag{49}$$

Both models therefore use an $O(KV)$ low-rank correction, and both can modulate the incoming value coordinatewise. Their factorizations differ:

Model	outer write direction	erased read direction	value multiplier
GDN-2	$\boldsymbol{k}_t$	$\boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t$	$\boldsymbol{w}_t$
KDN-2	$\boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t$	$\boldsymbol{k}_t$	$\boldsymbol{\beta}_t$

For KDN-2, the same $\beta_{t,j}$ multiplies the observation write and the erasure of the predicted value in column j. This tie is forced by the Kalman innovation. GDN-2 instead predicts $\boldsymbol{b}_t^{\text{erase}}$ and $\boldsymbol{w}_t$ independently, deliberately decoupling how much old content is erased from how much new value is injected.

The two covariance factors explain the two visible parts of the KDN-2 update. The key factor $\widehat{\boldsymbol{P}}_t$ changes the outer write direction from $\boldsymbol{k}_t$ to $\boldsymbol{g}_t$, while the value factor $\widehat{\boldsymbol{B}}_t$ and observation noise determine the relative entries of $\boldsymbol{\beta}_t$. These gates are functions of the carried uncertainty state and therefore depend on the history, not only on the current token representation.

4.2 Exact intersection of the update families

The similarity between Equations (46) and (49) does not imply that either family contains the other. Their common intersection is much smaller.

Proposition 4 (Algebraic GDN-2–KDN-2 intersection)**.** *At the level of the memory-update equations, assume a nondegenerate update with $\boldsymbol{k}_t \neq \mathbf{0}$ and require Equations (46) and (49) to agree for arbitrary $\widehat{\boldsymbol{S}}_t$ and $\boldsymbol{v}_t$. Then there must be scalars $\lambda_t > 0$ and $\beta_t > 0$ such that*

$$\begin{aligned} \boldsymbol{g}_t &= \lambda_t \boldsymbol{k}_t, & \boldsymbol{\beta}_t &= \beta_t \mathbf{1}_V, \\ \boldsymbol{w}_t &= \lambda_t \beta_t \mathbf{1}_V, & \boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t &= \lambda_t \beta_t \boldsymbol{k}_t. \end{aligned} \tag{50}$$

Conversely, these conditions make the two updates identical.

Proof. Equality of the coefficients multiplying each arbitrary $v_{t,j}$ requires

$$\beta_{t,j} \boldsymbol{g}_t = w_{t,j} \boldsymbol{k}_t. \tag{51}$$

For a nonzero write this forces $\boldsymbol{g}_t = \lambda_t \boldsymbol{k}_t$ and $w_{t,j} = \lambda_t \beta_{t,j}$. Equality of the linear maps applied to each arbitrary memory column then requires

$$\lambda_t \beta_{t,j} \boldsymbol{k}_t \boldsymbol{k}_t^{\mathsf{T}} = \boldsymbol{k}_t (\boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t)^{\mathsf{T}} \qquad \text{for every } j. \tag{52}$$

The right-hand side is shared across value columns, so every $\beta_{t,j}$ must equal one scalar β_t, and $\boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t = \lambda_t \beta_t \boldsymbol{k}_t$. Substitution proves sufficiency. □

For the implemented default GDN-2 parameterization, the converse additionally requires $\lambda_t \beta_t \in (0, 1)$ so that both gates lie in their sigmoid ranges. The optional negative-eigenvalue variant permits an erase coefficient up to two but leaves the write gate in $(0, 1)$.

For a dense key and diagonal $\widehat{\boldsymbol{P}}_t$, the condition $\boldsymbol{g}_t = \lambda_t \boldsymbol{k}_t$ requires $\widehat{p}_{t,i} = \lambda_t$ on the support of $\boldsymbol{k}_t$. The intersection therefore reduces to the scalar-gated delta update

$$\boldsymbol{S}_t = \widehat{\boldsymbol{S}}_t + \gamma_t \boldsymbol{k}_t (\boldsymbol{v}_t - \widehat{\boldsymbol{S}}_t^{\mathsf{T}} \boldsymbol{k}_t)^{\mathsf{T}}, \qquad \gamma_t = \lambda_t \beta_t, \tag{53}$$

which is the KDA form. Away from this intersection, KDN-2 can rotate the write direction through key-space uncertainty, which GDN-2 cannot express with its fixed outer direction $\boldsymbol{k}_t$; GDN-2 can independently choose erase and write gates, which a coherent KDN-2 innovation cannot express.

The one-axis boundaries expose the two separate mismatches. When $\widehat{\boldsymbol{P}}_t \propto \boldsymbol{I}_K$ but $\boldsymbol{\beta}_t$ remains value dependent, KDN-2 reduces to the value-wise isotropic update: it shares GDN-2's outer direction and value-wise write scaling, but still uses the same $\beta_{t,j}$ for both halves of each column's innovation. When $\widehat{\boldsymbol{B}}_t \propto \boldsymbol{I}_V$ and $\boldsymbol{R}_t \propto \boldsymbol{I}_V$, KDN-2 reduces to Diagonal KDN: its value gate becomes common, but the anisotropic direction $\widehat{\boldsymbol{P}}_t \boldsymbol{k}_t$ generally cannot be represented by GDN-2.

4.3 Interpretation and controlled comparisons

GDN-2 may be viewed as a discriminative relaxation of an uncertainty-gated delta rule. It predicts the two sides of the update directly and gains token-local flexibility, but it does not carry a covariance whose predictive variance or posterior contraction constrains those gates. KDN-2 instead derives its update from the separable predictive covariance and diagonal observation noise. Its restrictions provide a calibrated predictive distribution and a causal notion of future protection, at the cost of the coupled covariance projection derived above.

The connection suggests four controlled ablations:

1. **Native scalar innovation tying.** Within the existing GDN-2 operator, constrain $\boldsymbol{b}_t^{\text{erase}} = \gamma_t \mathbf{1}_K$ and $\boldsymbol{w}_t = \gamma_t \mathbf{1}_V$. This tests the benefit of decoupling while retaining the native shared erase system.
2. **Value-wise tied innovation.** Replace the shared erase operator by $-\boldsymbol{k}_t \boldsymbol{k}_t^\top \widehat{\boldsymbol{S}}_t \operatorname{diag}(\boldsymbol{\beta}_t)$ and use the same $\boldsymbol{\beta}_t$ on the incoming value. This is a structural recurrence ablation, not merely a gate constraint on GDN-2.
3. **Gain direction.** Replace $\boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t$ by a collinear direction

$$\boldsymbol{g}_{t,\|} = \frac{\boldsymbol{k}_t^\top \boldsymbol{g}_t}{\|\boldsymbol{k}_t\|_2^2} \boldsymbol{k}_t,$$

   which preserves $\boldsymbol{k}_t^\top \boldsymbol{g}_t$ and hence the current-key contraction while removing the off-key component.
4. **Stateful versus direct gates.** Match instantaneous write strengths while comparing uncertainty-derived gates with gates predicted only from the current token.

Useful diagnostics are the cosine between $\boldsymbol{g}_t$ and $\boldsymbol{k}_t$, dispersion across $\beta_{t,j}$, disagreement between GDN-2 erase and write strengths, predictive innovation NLL, and repeated-key protection.

5 Parallelism and approximation choices

The value-wise model in Section 2 has independent scalar information recurrences, but its value-dependent gains require separate memory systems. KDN-2 adds a second difficulty: its covariance projection couples all key and value factors at every token. Three implementation levels should be distinguished:

1. **Sequential joint projection.** Run the exact gain and several alternating reverse-KL sweeps per token. This is the reference semantics.
2. **Grouped values.** Share one b and one gain across a small value group. This reduces the number of memory systems while retaining coarse value reliability.
3. **Chunk-frozen factors.** Freeze or lag one factor within a chunk, use the corresponding one-sided update, and commit a joint projection at the boundary. This may recover useful matrix operations, but it is an approximation whose error must be measured against the sequential reference.

The compatible-noise construction in Equation (36) restores a shared gain, but it also removes the value factor from the mean and is therefore a control rather than a full KDN-2 implementation. Appendix A derives the exact value-batched compact-WY system for the value-wise model and makes explicit where its computation ceases to be shared across value coordinates. Appendix B then derives an exact two-stage implementation of two-axis KDN-2: a causal coupled-factor prepass followed by a value-batched chunkwise mean pass.

6 Parameterization, gauges, and validation

Positive factors. Let $\boldsymbol{h}_t$ denote the observed token representation supplied to the uncertainty heads. Predict diagonal process increments and observation noise through positive floors, for example

$$\boldsymbol{\omega}_t^{(k)} = \omega_{k,\min} + \operatorname{softplus}(f_k(\boldsymbol{h}_t)), \quad \boldsymbol{\omega}_t^{(v)} = \omega_{v,\min} + \operatorname{softplus}(f_v(\boldsymbol{h}_t)), \quad \boldsymbol{r}_t = r_{\min} + \operatorname{softplus}(f_r(\boldsymbol{h}_t)). \tag{54}$$

Fixing $\boldsymbol{r}_t = r\mathbf{1}_V$ while learning value-specific $\boldsymbol{\omega}_t^{(v)}$ is a useful first experiment: it breaks value symmetry without learning the most direct $b_{t,j}/r_{t,j}$ confound. In addition to the exact internal factor gauge in Equation (21), jointly scaling the full covariance and observation noise produces the usual Kalman covariance-scale gauge.

Symmetry breaking. Initialize $\boldsymbol{B}_0 = \boldsymbol{I}_V$ and $\boldsymbol{P}_0 = \boldsymbol{I}_K$ for a neutral gauge. Value uncertainty remains tied unless the initial value factor, value-process uncertainty, or observation noise varies across value coordinates. Realized values alone cannot change covariance symmetry.

Required tests. A KDN-2 implementation should pass the following checks before language-model training:

1. Compare the exact mean and covariance in Equations (31) and (34) against a dense Kalman update on small random systems.
2. Verify the Diagonal-KDN and value-wise isotropic limits in Equations (44) and (45).
3. Check factor-gauge invariance numerically under $(\boldsymbol{B}, \boldsymbol{P}) \mapsto (\lambda \boldsymbol{B}, \boldsymbol{P}/\lambda)$.
4. Confirm that the alternating projection satisfies the row- and column-sum conditions in Equation (41) and decreases the reverse-KL objective.
5. Compare zero, one, two, and converged projection sweeps; report covariance error, gain error at later tokens, throughput, and peak memory.
6. Test repeated keys, orthogonal keys with identical elementwise squares, value-specific corruption, and value-basis rotations. The last test exposes the basis dependence of a diagonal value factor.

7 Limitations and experimental contract

KDN-2 represents a rank-one factorization of the $K \times V$ table of entrywise variances. It is strictly richer than either one-axis family but much smaller than a fully independent diagonal covariance with KV states. It still assumes no cross-key or cross-value posterior covariance, depends on the learned key and value bases, and introduces both an exact factor gauge and strong confounding between value variance and observation noise.

The most important limitation is computational rather than notational. The exact posterior mean is inexpensive, but the joint covariance projection is coupled and the value-dependent gains break the shared memory transition. A claimed KDN-2 kernel must therefore specify which projection it implements, how many alternating sweeps it uses, how the gauge is fixed, and whether the memory system is value-wise, grouped, or chunk-approximated. Results should be compared with the two one-sided boundaries at matched parameter count and effective write strength.

The decisive initial experiment is a small synthetic ablation with four arms:

$$\begin{array}{ll} \text{Diagonal KDN,} & \text{value-wise isotropic KDN,} \\ \text{KDN-2 with one projection sweep,} & \text{KDN-2 with a converged projection.} \end{array} \tag{55}$$

This separates the benefit of two-axis uncertainty from the cost and approximation error of the joint projection before committing to a production kernel.

8 Conclusion

Value-wise isotropic uncertainty and Diagonal KDN allocate the same kind of scalar uncertainty budget to opposite axes of the memory matrix. KDN-2 combines them through the separable covariance $\boldsymbol{B}_t \otimes \boldsymbol{P}_t$. Its exact mean update cleanly factors into a key-wise uncertainty direction and value-wise uncertainty gates. Its covariance update does not factor cleanly: useful value-dependent gains destroy Kronecker closure, and the principled reverse-KL projection is a coupled matrix-balancing problem. This distinction is the central design constraint. KDN-2 is mathematically well defined as a sequential filter and a useful research model, but a scan-parallel approximation must be stated and validated separately.

A Chunkwise implementation of value-wise isotropic KDN

This appendix derives an exact chunkwise evaluation of the value-wise isotropic model in Section 2. The derivation follows the compact-WY construction for the delta rule: first remove the diagonal decay, then expose the causal residuals as a unit-lower triangular system, and finally express all reads and the outgoing state as matrix products. The only structural change is important: every value coordinate has its own gain sequence, so the triangular system is value batched rather than shared.

A.1 Two-stage scan

The uncertainty state does not depend on the realized values or on the retained memory. Hence it can be evaluated before the memory recurrence. For each value coordinate j, write the information state homogeneously as

$$\boldsymbol{z}_{t,j}^{(c)} = \begin{bmatrix} n_{t,j} \\ d_{t,j} \end{bmatrix}, \qquad c_{t,j} = \frac{n_{t,j}}{d_{t,j}}, \qquad \boldsymbol{z}_{t,j}^{(c)} = \boldsymbol{M}_{t,j} \boldsymbol{z}_{t-1,j}^{(c)}, \tag{56}$$

where $\boldsymbol{M}_{t,j}$ is the 2×2 matrix in Equation (17). Products of these matrices are associative. A chunked reduce–scan–replay pass therefore produces every exclusive $c_{t-1,j}$ and hence every $\widehat{b}_{t,j}$ and $\beta_{t,j}$ before the memory pass. Explicitly, the exclusive state starts from $\boldsymbol{z}_{0,j}^{(c)} = [c_{0,j}, 1]^\mathsf{T}$ and emits

$$\widehat{c}_{t,j} = \frac{c_{t-1,j}}{a_t + \omega_{t,j} c_{t-1,j}}, \qquad \beta_{t,j} = \frac{1}{r_{t,j}\widehat{c}_{t,j} + \|\boldsymbol{k}_t\|_2^2}, \tag{57}$$

before $\boldsymbol{M}_{t,j}$ incorporates token t. In homogeneous coordinates, the same emission can avoid intermediate divisions: with $\chi_{t,j} = a_t d_{t-1,j} + \omega_{t,j} n_{t-1,j}$,

$$\widehat{c}_{t,j} = \frac{n_{t-1,j}}{\chi_{t,j}}, \qquad \beta_{t,j} = \frac{\chi_{t,j}}{r_{t,j} n_{t-1,j} + \|\boldsymbol{k}_t\|_2^2 \chi_{t,j}}. \tag{58}$$

Homogeneous vectors and chunk summaries may be rescaled after multiplication without changing their represented precision. The gain and memory chunk lengths need not be the same. What follows conditions on these precomputed gains.

Consider one memory chunk of length C, use local indices $r = 1, \ldots, C$, and suppress its global chunk index. Let $\boldsymbol{S}^{\text{in}} = \boldsymbol{S}_0$ be the incoming memory. Substituting $\widehat{\boldsymbol{S}}_r = \boldsymbol{D}_r \boldsymbol{S}_{r-1}$ into Equation (12) gives

$$\boldsymbol{S}_r = \boldsymbol{D}_r \boldsymbol{S}_{r-1} + \boldsymbol{k}_r \left[\boldsymbol{\beta}_r \odot \left(\boldsymbol{v}_r - (\boldsymbol{D}_r \boldsymbol{S}_{r-1})^\mathsf{T} \boldsymbol{k}_r\right)\right]^\mathsf{T}. \tag{59}$$

A.2 Removing the diagonal decay

Assume the gated transition has $\alpha_{r,i} > 0$. Define the within-chunk cumulative decay

$$\boldsymbol{\gamma}_r = \prod_{\ell=1}^{r} \boldsymbol{\alpha}_\ell, \qquad \boldsymbol{\Gamma}_r = \operatorname{diag}(\boldsymbol{\gamma}_r), \qquad \boldsymbol{\gamma}_0 = \mathbf{1}, \tag{60}$$

where the product is elementwise. Since $\boldsymbol{D}_r \boldsymbol{\Gamma}_{r-1} = \boldsymbol{\Gamma}_r$, write

$$\boldsymbol{S}_r = \boldsymbol{\Gamma}_r \boldsymbol{Y}_r, \qquad \overline{\boldsymbol{k}}_r = \boldsymbol{\Gamma}_r \boldsymbol{k}_r = \boldsymbol{\gamma}_r \odot \boldsymbol{k}_r, \qquad \overline{\boldsymbol{p}}_r = \boldsymbol{\Gamma}_r^{-1} \boldsymbol{k}_r = \boldsymbol{k}_r \oslash \boldsymbol{\gamma}_r. \tag{61}$$

The incoming normalized state is $\boldsymbol{Y}_0 = \boldsymbol{S}^{\text{in}}$. Multiplying Equation (59) by $\boldsymbol{\Gamma}_r^{-1}$ gives the undecayed two-key delta rule

$$\boldsymbol{Y}_r = \boldsymbol{Y}_{r-1} + \overline{\boldsymbol{p}}_r \left[\boldsymbol{\beta}_r \odot \left(\boldsymbol{v}_r - \boldsymbol{Y}_{r-1}^\top \overline{\boldsymbol{k}}_r \right) \right]^\top. \tag{62}$$

Thus $\overline{\boldsymbol{k}}_r$ is the read key and $\beta_{r,j} \overline{\boldsymbol{p}}_r$ is the write key for value coordinate j. Implementations should form the required decay ratios from cumulative log decays rather than explicitly dividing by a potentially small $\boldsymbol{\gamma}_r$.

A.3 Causal innovations as value-batched triangular systems

Stack the normalized read and base-write keys by row,

$$[\overline{\boldsymbol{K}}]_{r,:} = \overline{\boldsymbol{k}}_r^\top, \qquad [\overline{\boldsymbol{P}}]_{r,:} = \overline{\boldsymbol{p}}_r^\top, \qquad \boldsymbol{L} = \operatorname{tril}\left(\overline{\boldsymbol{K}} \, \overline{\boldsymbol{P}}^\top, -1 \right). \tag{63}$$

The strictly lower matrix $\boldsymbol{L} \in \mathbb{R}^{C \times C}$ is shared by all value coordinates. For coordinate j, define

$$\begin{aligned}
&\boldsymbol{v}_j = (v_{1,j}, \ldots, v_{C,j})^\top, \quad \boldsymbol{\Lambda}_j^{(\beta)} = \operatorname{diag}(\beta_{1,j}, \ldots, \beta_{C,j}), \\
&\varepsilon_{r,j} = v_{r,j} - \overline{\boldsymbol{k}}_r^\top \boldsymbol{y}_{r-1,j}, \qquad \boldsymbol{\varepsilon}_j = (\varepsilon_{1,j}, \ldots, \varepsilon_{C,j})^\top, \\
&\boldsymbol{\xi}_j = \boldsymbol{\Lambda}_j^{(\beta)} \boldsymbol{\varepsilon}_j.
\end{aligned} \tag{64}$$

Unrolling only the preceding writes in Equation (62) yields

$$\boldsymbol{\varepsilon}_j = \boldsymbol{v}_j - \overline{\boldsymbol{K}} \boldsymbol{s}_j^{\text{in}} - \boldsymbol{L} \boldsymbol{\xi}_j, \qquad \boldsymbol{\xi}_j = \boldsymbol{\Lambda}_j^{(\beta)} \boldsymbol{\varepsilon}_j. \tag{65}$$

Consequently, the complete set of causal innovations in the chunk is the unique solution of either equivalent unit-lower system

$$\begin{aligned}
(\boldsymbol{I}_C + \boldsymbol{L} \boldsymbol{\Lambda}_j^{(\beta)}) \boldsymbol{\varepsilon}_j &= \boldsymbol{v}_j - \overline{\boldsymbol{K}} \boldsymbol{s}_j^{\text{in}}, \\
(\boldsymbol{I}_C + \boldsymbol{\Lambda}_j^{(\beta)} \boldsymbol{L}) \boldsymbol{\xi}_j &= \boldsymbol{\Lambda}_j^{(\beta)} (\boldsymbol{v}_j - \overline{\boldsymbol{K}} \boldsymbol{s}_j^{\text{in}}).
\end{aligned} \tag{66}$$

The first form remains well defined when a gain is zero and is preferable to writing a system with $(\boldsymbol{\Lambda}_j^{(\beta)})^{-1}$. Let $\boldsymbol{\Xi} = [\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_V] \in \mathbb{R}^{C \times V}$, with $\boldsymbol{\xi}_r^\top$ denoting row r. Equivalently, all value coordinates can be advanced together by the fused forward substitution

$$\boldsymbol{\xi}_r = \boldsymbol{\beta}_r \odot \left(\boldsymbol{v}_r - (\boldsymbol{S}^{\text{in}})^\top \overline{\boldsymbol{k}}_r - \sum_{i<r} L_{ri} \boldsymbol{\xi}_i \right), \qquad \boldsymbol{\xi}_r \in \mathbb{R}^V. \tag{67}$$

This form makes the reusable geometry $\boldsymbol{L}$ and the elementwise value-axis gain explicit.

To recover the usual compact-WY notation, define

$$\boldsymbol{A}_j = (\boldsymbol{I}_C + \boldsymbol{L}\boldsymbol{\Lambda}_j^{(\beta)})^{-1}, \qquad \boldsymbol{\eta}_j = \boldsymbol{A}_j \boldsymbol{v}_j, \qquad \boldsymbol{W}_j = \boldsymbol{A}_j \overline{\boldsymbol{K}}. \tag{68}$$

Then

$$\boldsymbol{\varepsilon}_j = \boldsymbol{\eta}_j - \boldsymbol{W}_j \boldsymbol{s}_j^{\text{in}}, \qquad \boldsymbol{\xi}_j = \boldsymbol{\Lambda}_j^{(\beta)}(\boldsymbol{\eta}_j - \boldsymbol{W}_j \boldsymbol{s}_j^{\text{in}}). \tag{69}$$

The inverse in Equation (68) is notation: an implementation uses a unit-lower-triangular solve. The factors $\boldsymbol{\eta}_j$ and $\boldsymbol{W}_j$ depend only on inputs from the current chunk and may therefore be prepared for all chunks in parallel, before the incoming chunk states are known.

The connection to the compact-WY product is explicit. Define the normalized rank-one transition

$$\boldsymbol{E}_{r,j} = \boldsymbol{I}_K - \beta_{r,j} \overline{\boldsymbol{p}}_r \overline{\boldsymbol{k}}_r^{\top}. \tag{70}$$

Then the product of all within-chunk transitions is

$$\boldsymbol{E}_{C,j} \boldsymbol{E}_{C-1,j} \cdots \boldsymbol{E}_{1,j} = \boldsymbol{I}_K - \overline{\boldsymbol{P}}^{\top} \boldsymbol{\Lambda}_j^{(\beta)} \boldsymbol{W}_j. \tag{71}$$

The same transformed residuals also collect all forced writes, so

$$\boldsymbol{y}_{C,j} = \left(\boldsymbol{I}_K - \overline{\boldsymbol{P}}^{\top} \boldsymbol{\Lambda}_j^{(\beta)} \boldsymbol{W}_j\right) \boldsymbol{s}_j^{\text{in}} + \overline{\boldsymbol{P}}^{\top} \boldsymbol{\Lambda}_j^{(\beta)} \boldsymbol{\eta}_j = \boldsymbol{s}_j^{\text{in}} + \overline{\boldsymbol{P}}^{\top} \boldsymbol{\xi}_j. \tag{72}$$

A.4 Parallel reads and the outgoing state

The normalized state after any local prefix r is

$$\boldsymbol{Y}_r = \boldsymbol{S}^{\text{in}} + \sum_{i=1}^{r} \overline{\boldsymbol{p}}_i \boldsymbol{\xi}_i^{\top}, \tag{73}$$

where $\boldsymbol{\xi}_i^{\top}$ is row i of $\boldsymbol{\Xi}$. In particular, the outgoing physical memory is

$$\boldsymbol{S}^{\text{out}} = \boldsymbol{\Gamma}_C \left(\boldsymbol{S}^{\text{in}} + \overline{\boldsymbol{P}}^{\top} \boldsymbol{\Xi}\right). \tag{74}$$

For completeness, suppose the mixer reads after the current write with queries $\boldsymbol{q}_r \in \mathbb{R}^K$ and outputs $\boldsymbol{o}_r = \boldsymbol{S}_r^{\top} \boldsymbol{q}_r$. Define

$$[\overline{\boldsymbol{Q}}]_{r,:} = (\boldsymbol{\Gamma}_r \boldsymbol{q}_r)^{\top}, \qquad \boldsymbol{L}_{qp} = \operatorname{tril}(\overline{\boldsymbol{Q}}\overline{\boldsymbol{P}}^{\top}). \tag{75}$$

The diagonal is retained because the read is post-write. All outputs in the chunk are

$$\boldsymbol{O} = \overline{\boldsymbol{Q}} \boldsymbol{S}^{\text{in}} + \boldsymbol{L}_{qp} \boldsymbol{\Xi}. \tag{76}$$

A pre-write read instead uses the strictly lower part of the same score matrix.

The inverse cumulative decay in $\overline{\boldsymbol{P}}$ need not be materialized. The entries used in the shared score matrices are

$$\begin{aligned} L_{ri} &= \boldsymbol{k}_r^{\top} \operatorname{diag}(\boldsymbol{\gamma}_r \oslash \boldsymbol{\gamma}_i) \boldsymbol{k}_i, \qquad && i < r, \\ (L_{qp})_{ri} &= \boldsymbol{q}_r^{\top} \operatorname{diag}(\boldsymbol{\gamma}_r \oslash \boldsymbol{\gamma}_i) \boldsymbol{k}_i, \qquad && i \le r. \end{aligned} \tag{77}$$

Likewise, define the tail-adjusted write matrix by

$$[\boldsymbol{K}_{\text{tail}}]_{r,:} = [(\boldsymbol{\gamma}_C \oslash \boldsymbol{\gamma}_r) \odot \boldsymbol{k}_r]^{\top}. \tag{78}$$

Equation (74) can then be evaluated as

$$\boldsymbol{S}^{\text{out}} = \boldsymbol{\Gamma}_C \boldsymbol{S}^{\text{in}} + \boldsymbol{K}_{\text{tail}}^{\mathsf{T}} \boldsymbol{\Xi}. \tag{79}$$

The ratios in Equations (77) and (78) contain only the decay between the write and the read or chunk boundary. They should be computed from cumulative log-decay differences. This remains stable even when the individual cumulative products underflow.

Proposition 5 (Exactness of the value-batched chunk form)**.** *For fixed gains $\{\boldsymbol{\beta}_r\}_{r=1}^{C}$ and positive diagonal decays, the solutions of Equation (66) and the prefix reconstruction in Equation (73) equal the token-by-token recurrence in Equation (59). Consequently, Equations (74) and (76) give its exact chunk boundary and post-write outputs.*

Proof. Equation (62) implies $\boldsymbol{y}_{r-1,j} = \boldsymbol{s}_j^{\text{in}} + \sum_{i<r} \overline{\boldsymbol{p}}_i \xi_{i,j}$. Substitution into the definition of $\varepsilon_{r,j}$ gives row r of Equation (65). Its coefficient matrix is unit lower triangular, so its solution is unique and equals the sequential innovations. Substituting those innovations back into the normalized recurrence gives Equation (73); rescaling and applying the queries gives Equations (74) and (76). □

A.5 Parallel schedule, cost, and sharing limit

An exact implementation has the following schedule:

1. Compute all $\beta_{t,j}$ with the independent Möbius scans in Section A.1.
2. For every memory chunk in parallel, form the cumulative decay, the normalized keys, and the shared score matrices $\boldsymbol{L}$ and $\boldsymbol{L}_{qp}$.
3. Solve Equation (66) batched over value coordinates, then evaluate Equations (76) and (74). Only the $K \times V$ memory is carried between chunks.

There are two useful scheduling choices. Precomputing $\boldsymbol{\eta}_j$ and $\boldsymbol{W}_j$ exposes the same short inter-chunk spine as the conventional compact-WY algorithm, but requires a distinct $C \times K$ factor $\boldsymbol{W}_j$ for every value coordinate. Forming these factors costs $O(VC^2K)$ work and storing them costs $O(VCK)$ per chunk. Alternatively, a kernel can retain the shared $\boldsymbol{L}$ and perform the one-right-hand-side solves only after $\boldsymbol{S}^{\text{in}}$ arrives. This avoids the value-wise $\boldsymbol{W}_j$ tensors and costs

$$O(C^2K + C^2V + CKV) \tag{80}$$

work per chunk, but moves the triangular solve onto the inter-chunk dependency path. A practical kernel can tile the value dimension, reuse each tile of $\boldsymbol{L}$, and use matrix products for the off-diagonal block updates. The reverse pass uses the corresponding value-batched unit-upper-triangular solves and has the same asymptotic work; chunk-local factors may be recomputed rather than saved.

The matrices $\boldsymbol{L}$ and $\boldsymbol{L}_{qp}$ are shared, but the actual WY operator is not:

$$\boldsymbol{A}_j = (\boldsymbol{I}_C + \boldsymbol{L}\boldsymbol{\Lambda}_j^{(\beta)})^{-1}. \tag{81}$$

Thus arbitrary value-dependent gains require V batched triangular systems. If $\beta_{r,j} = \beta_r$ for every j, all $\boldsymbol{\Lambda}_j^{(\beta)}$ coincide and one shared WY solve with V right-hand sides recovers the standard KDA/DeltaNet implementation. Sharing a gain within each of G value groups similarly requires only G systems; it is exact for the grouped model and an approximation to the fully value-wise model.

Finally, this appendix parallelizes the value-wise isotropic boundary, for which the gain stage is itself a collection of scalar Möbius scans. Conditional on known $\boldsymbol{g}_t$ and $\boldsymbol{\beta}_t$, the KDN-2 mean admits the same two-key derivation after replacing $\overline{\boldsymbol{p}}_t$ by $\boldsymbol{\Gamma}_t^{-1}\boldsymbol{g}_t$. The joint KDN-2 covariance projection, however, still couples the key and value factors from one token to the next, so this conditional WY construction does not by itself make the full KDN-2 filter scan parallel. Here "exact" means exact evaluation of the retained value-wise recurrence, including the trace-projected covariance prediction in Equation (8); it does not turn that projected family into the unrestricted dense Kalman filter.

B Two-stage chunkwise implementation of two-axis KDN-2

This appendix derives an exact two-stage implementation of the retained two-axis KDN-2 recurrence in Section 3. The first stage advances the coupled key–value uncertainty factors and emits the coefficients required by the mean update. The second stage conditions on those coefficients and applies the two-key compact-WY construction. The distinction is essential: the mean pass is chunkwise, whereas the joint reverse-KL factor pass remains causal and sequential.

Here "exact" means equality with the token-by-token retained KDN-2 recurrence under the same prediction, factor gauge, and reverse-KL projection—not equality with the unrestricted dense Kalman filter.

B.1 Stage 1: causal coupled-factor prepass

The covariance dynamics do not depend on the realized values or on the memory mean. They may therefore be evaluated before the $K \times V$ mean recurrence. Starting from the projected factors $(\boldsymbol{P}_{t-1}, \boldsymbol{B}_{t-1})$, compute

$$\widehat{\boldsymbol{P}}_t = \boldsymbol{D}_t \boldsymbol{P}_{t-1} \boldsymbol{D}_t^{\top} + \boldsymbol{\Omega}_t^{(k)}, \quad \widehat{\boldsymbol{B}}_t = \boldsymbol{B}_{t-1} + \boldsymbol{\Omega}_t^{(v)}. \tag{82}$$

Fix the predictive factor gauge, for example by enforcing $\det(\widehat{\boldsymbol{B}}_t) = 1$ and applying the compensating inverse scale to $\widehat{\boldsymbol{P}}_t$. The exclusive predictive factors emit

$$\begin{aligned} \boldsymbol{g}_t &= \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t, & z_t &= \boldsymbol{k}_t^{\top} \boldsymbol{g}_t, \\ \Delta_{t,j} &= r_{t,j} + \widehat{b}_{t,j} z_t, & \beta_{t,j} &= \frac{\widehat{b}_{t,j}}{\Delta_{t,j}}. \end{aligned} \tag{83}$$

These quantities are emitted before projecting the current posterior: the current mean update uses the exclusive factors $\widehat{\boldsymbol{P}}_t$ and $\widehat{\boldsymbol{B}}_t$, not the post-write factors.

After this emission, initialize the candidate posterior at the predictive factors and solve the coupled reverse-KL projection. One alternating sweep computes

$$\begin{aligned} A_P &= \sum_{i=1}^{K} \frac{p_{t,i}}{\widehat{p}_{t,i}}, \quad Z_P = \sum_{i=1}^{K} p_{t,i} k_{t,i}^2, \quad b_{t,j} \leftarrow \frac{K}{A_P/\widehat{b}_{t,j} + Z_P/r_{t,j}}, \\ A_B &= \sum_{j=1}^{V} \frac{b_{t,j}}{\widehat{b}_{t,j}}, \quad T_B = \sum_{j=1}^{V} \frac{b_{t,j}}{r_{t,j}}, \qquad p_{t,i} \leftarrow \frac{V}{A_B/\widehat{p}_{t,i} + T_B k_{t,i}^2}. \end{aligned} \tag{84}$$

Gauge normalization follows each complete sweep. Iteration to convergence gives the unique reverse-KL optimum in the fixed gauge. Alternatively, a fixed number N_{proj} of sweeps defines a truncated-projection KDN-2 recurrence. In either case, the resulting $(\boldsymbol{P}_t, \boldsymbol{B}_t)$ are carried to token $t+1$, while $\boldsymbol{g}_t$, $\boldsymbol{\beta}_t$, and optionally $\boldsymbol{\Delta}_t$ are stored for Stage 2.

Exact scalarization of the coupled projection. The apparent $(K+V)$-dimensional fixed point has a one-dimensional representation because the balancing matrix in Equation (41) is the sum of two rank-one terms. Define the predictive axis statistics and factor-summary ratios

$$\zeta_{t,i}^{(k)} = \widehat{p}_{t,i} k_{t,i}^2, \qquad \zeta_{t,j}^{(v)} = \frac{\widehat{b}_{t,j}}{r_{t,j}}, \qquad \tau_t^{(k)} = \frac{Z_P}{A_P}, \qquad \tau_t^{(v)} = \frac{T_B}{A_B}. \tag{85}$$

Let $\boldsymbol{\zeta}_t^{(k)} \in \mathbb{R}_+^K$ and $\boldsymbol{\zeta}_t^{(v)} \in \mathbb{R}_+^V$ collect the corresponding axis statistics. For a nonnegative vector $\boldsymbol{a} \in \mathbb{R}_+^m$, introduce

$$D_{\boldsymbol{a}}(s) = \sum_{\ell=1}^{m} \frac{1}{1+sa_\ell}, \quad N_{\boldsymbol{a}}(s) = \sum_{\ell=1}^{m} \frac{a_\ell}{1+sa_\ell}, \quad F_{\boldsymbol{a}}(s) = \frac{N_{\boldsymbol{a}}(s)}{D_{\boldsymbol{a}}(s)}. \tag{86}$$

The stationary equations in Equation (84) imply

$$\frac{b_{t,j}}{\widehat{b}_{t,j}} = \frac{K/A_P}{1+\tau_t^{(k)}\zeta_{t,j}^{(v)}}, \qquad \frac{p_{t,i}}{\widehat{p}_{t,i}} = \frac{V/A_B}{1+\tau_t^{(v)}\zeta_{t,i}^{(k)}}. \tag{87}$$

Taking the defining ratios reduces the joint projection to

$$\tau_t^{(v)} = F_{\boldsymbol{\zeta}_t^{(v)}}(\tau_t^{(k)}), \qquad \tau_t^{(k)} = F_{\boldsymbol{\zeta}_t^{(k)}}(\tau_t^{(v)}), \qquad \Longleftrightarrow \qquad \tau_t^{(k)} = F_{\boldsymbol{\zeta}_t^{(k)}}\left(F_{\boldsymbol{\zeta}_t^{(v)}}(\tau_t^{(k)})\right). \tag{88}$$

Thus the converged reverse-KL projection requires one scalar root rather than alternating writes of all $K+V$ candidate factors. Define $G_t(s) = F_{\boldsymbol{\zeta}_t^{(k)}}(F_{\boldsymbol{\zeta}_t^{(v)}}(s))$ and $\mathcal{R}_t(s) = G_t(s) - s$. Since G_t is a weighted mean of the entries of $\boldsymbol{\zeta}_t^{(k)}$,

$$\mathcal{R}_t\left(\min_i \zeta_{t,i}^{(k)}\right) \geq 0, \qquad \mathcal{R}_t\left(\max_i \zeta_{t,i}^{(k)}\right) \leq 0. \tag{89}$$

Uniqueness follows from the strict convexity of the gauge-fixed reverse-KL objective. If all $\zeta_{t,i}^{(k)}$ equal one constant, that constant is the root directly; otherwise safeguarded Newton or bisection applies on the bracket above. For deterministic GPU execution, use a fixed maximum iteration count and the gauge-invariant relative residual $|G_t(s) - s|/(G_t(s)+s) \leq \epsilon_{\mathrm{root}}$, with the all-zero case handled directly.

After solving for $\tau_t^{(k)}$ and setting $\tau_t^{(v)} = F_{\boldsymbol{\zeta}_t^{(v)}}(\tau_t^{(k)})$, one convenient raw gauge is

$$p_{t,i}^{(0)} = \frac{\widehat{p}_{t,i}}{1+\tau_t^{(v)}\zeta_{t,i}^{(k)}}, \qquad b_{t,j}^{(0)} = \frac{(1+\tau_t^{(k)}\tau_t^{(v)})\widehat{b}_{t,j}}{1+\tau_t^{(k)}\zeta_{t,j}^{(v)}}. \tag{90}$$

Every gauge-equivalent solution is $p_{t,i} = p_{t,i}^{(0)}/\lambda_t$ and $b_{t,j} = \lambda_t b_{t,j}^{(0)}$. For the determinant gauge in Equation (22), choose

$$\lambda_t = \exp\left(-\frac{1}{V}\sum_{j=1}^{V} \log b_{t,j}^{(0)}\right). \tag{91}$$

The gauge-free entry variances are

$$p_{t,i}b_{t,j} = \frac{(1+\tau_t^{(k)}\tau_t^{(v)})\widehat{p}_{t,i}\widehat{b}_{t,j}}{(1+\tau_t^{(v)}\zeta_{t,i}^{(k)})(1+\tau_t^{(k)}\zeta_{t,j}^{(v)})}. \tag{92}$$

The identity $D_{\boldsymbol{a}}(s)+sN_{\boldsymbol{a}}(s)=m$ verifies the fixed-point form of Equation (84) by direct substitution.

The scalar form also preserves a deliberately truncated projection. Predictive initialization gives

$$\tau_t^{(k,0)}=\frac{1}{K}\sum_{i=1}^{K}\zeta_{t,i}^{(k)},\qquad \tau_t^{(v,n)}=F_{\boldsymbol{\zeta}_t^{(v)}}(\tau_t^{(k,n)}),\qquad \tau_t^{(k,n+1)}=F_{\boldsymbol{\zeta}_t^{(k)}}(\tau_t^{(v,n)}). \tag{93}$$

Here $\tau_t^{(k,n)}$ is the incoming p-summary used by the b update in sweep n, whereas $\tau_t^{(k,n+1)}$ is the outgoing p-summary. For $N_{\mathrm{proj}}\geq 1$, the last outgoing summary need not be evaluated: use the lagged pair $(\tau_t^{(k,N_{\mathrm{proj}}-1)},\tau_t^{(v,N_{\mathrm{proj}}-1)})$ in Equation (90). Indeed, $D_{\boldsymbol{\zeta}_t^{(v)}}(s)+sN_{\boldsymbol{\zeta}_t^{(v)}}(s)=V$ and $\tau^{(v)}=N_{\boldsymbol{\zeta}_t^{(v)}}(s)/D_{\boldsymbol{\zeta}_t^{(v)}}(s)$ imply $V/D_{\boldsymbol{\zeta}_t^{(v)}}(s)=1+s\tau^{(v)}$, precisely the reconstruction prefactor. The result is therefore the same posterior covariance, up to gauge, as N_{proj} complete b-then-p sweeps, not a new approximation. The case $N_{\mathrm{proj}}=0$ simply returns the predictive factors.

On a GPU, $\boldsymbol{\zeta}_t^{(k)}$ and $\boldsymbol{\zeta}_t^{(v)}$ can remain on chip while each scalar iteration performs one reduction along each axis. The vectors $\boldsymbol{p}_t$ and $\boldsymbol{b}_t$ are reconstructed and gauge-normalized only once per token. A persistent block for each independent batch–head pair can fuse prediction, gain emission, scalar projection, reconstruction, and gauge normalization. For a converged root, implicit differentiation can avoid saving or unrolling the root iterations; a fixed truncated solver should instead be differentiated through to preserve its specified recurrence. These changes can reduce synchronization and do reduce factor-vector traffic inside a token, but they do not remove the causal dependence between tokens.

Why Stage 1 is not an associative scan. The map

$$(\boldsymbol{P}_{t-1},\boldsymbol{B}_{t-1})\longmapsto(\boldsymbol{P}_t,\boldsymbol{B}_t) \tag{94}$$

contains a joint nonlinear matrix-scaling solve whose summaries depend on both candidate factors. Unlike the 2×2 Möbius matrices of the one-axis recurrence, composing token maps does not produce another token-local map in the same $O(K+V)$-parameter family. Stage 1 chunks must therefore be traversed in causal order; chunking provides tiling and checkpointing, not time parallelism.

B.2 Stage 2: conditional two-key mean recurrence

Once Stage 1 has produced $\boldsymbol{g}_t$ and $\boldsymbol{\beta}_t$, the memory recurrence is

$$\boldsymbol{S}_t=\boldsymbol{D}_t\boldsymbol{S}_{t-1}+\boldsymbol{g}_t\left[\boldsymbol{\beta}_t\odot\left(\boldsymbol{v}_t-(\boldsymbol{D}_t\boldsymbol{S}_{t-1})^{\mathsf{T}}\boldsymbol{k}_t\right)\right]^{\mathsf{T}}. \tag{95}$$

Consider one chunk of length C, relabel its tokens by $r=1,\dots,C$, and let $\boldsymbol{S}^{\mathrm{in}}=\boldsymbol{S}_0$ denote its incoming physical memory. Assume positive diagonal decays and define

$$\boldsymbol{\Gamma}_r=\boldsymbol{D}_r\boldsymbol{D}_{r-1}\cdots\boldsymbol{D}_1,\qquad \boldsymbol{\Gamma}_0=\boldsymbol{I}_K,\qquad \boldsymbol{S}_r=\boldsymbol{\Gamma}_r\boldsymbol{Y}_r. \tag{96}$$

The decay-scaled read and base-write directions are

$$\boldsymbol{k}_r^{+}=\boldsymbol{\Gamma}_r^{\mathsf{T}}\boldsymbol{k}_r,\qquad \boldsymbol{g}_r^{-}=\boldsymbol{\Gamma}_r^{-1}\boldsymbol{g}_r. \tag{97}$$

Since $\boldsymbol{D}_r\boldsymbol{\Gamma}_{r-1}=\boldsymbol{\Gamma}_r$, multiplying Equation (95) by $\boldsymbol{\Gamma}_r^{-1}$ gives

$$\boldsymbol{Y}_r=\boldsymbol{Y}_{r-1}+\boldsymbol{g}_r^{-}\left[\boldsymbol{\beta}_r\odot\left(\boldsymbol{v}_r-\boldsymbol{Y}_{r-1}^{\mathsf{T}}\boldsymbol{k}_r^{+}\right)\right]^{\mathsf{T}},\qquad \boldsymbol{Y}_0=\boldsymbol{S}^{\mathrm{in}}. \tag{98}$$

Thus normalized KDN-2 is a two-key delta rule: $\boldsymbol{k}_r^{+}$ reads, while $\beta_{r,j}\boldsymbol{g}_r^{-}$ writes value coordinate j.

B.3 Value-batched causal systems and compact-WY form

Stack the scaled directions by row,

$$[\boldsymbol{K}_+]_{r,:} = (\boldsymbol{k}_r^+)^\top, \qquad [\boldsymbol{G}_-]_{r,:} = (\boldsymbol{g}_r^-)^\top, \qquad \boldsymbol{L} = \operatorname{tril}(\boldsymbol{K}_+\boldsymbol{G}_-^\top, -1). \tag{99}$$

The strictly lower interaction matrix $\boldsymbol{L} \in \mathbb{R}^{C\times C}$ is shared across value coordinates. For column j, define

$$\begin{aligned} \boldsymbol{v}_j &= (v_{1,j}, \dots, v_{C,j})^\top, \qquad \boldsymbol{\Lambda}_j = \operatorname{diag}(\beta_{1,j}, \dots, \beta_{C,j}), \\ \varepsilon_{r,j} &= v_{r,j} - (\boldsymbol{k}_r^+)^\top \boldsymbol{y}_{r-1,j}, \quad \boldsymbol{\xi}_j = \boldsymbol{\Lambda}_j \boldsymbol{\varepsilon}_j. \end{aligned} \tag{100}$$

Unrolling only writes from earlier positions gives

$$\boldsymbol{\varepsilon}_j = \boldsymbol{v}_j - \boldsymbol{K}_+\boldsymbol{s}_j^{\text{in}} - \boldsymbol{L}\boldsymbol{\xi}_j, \qquad \boldsymbol{\xi}_j = \boldsymbol{\Lambda}_j\boldsymbol{\varepsilon}_j. \tag{101}$$

Therefore the causal innovations are the unique solutions of either equivalent unit-lower system

$$\begin{aligned} (\boldsymbol{I}_C + \boldsymbol{L}\boldsymbol{\Lambda}_j)\boldsymbol{\varepsilon}_j &= \boldsymbol{v}_j - \boldsymbol{K}_+\boldsymbol{s}_j^{\text{in}}, \\ (\boldsymbol{I}_C + \boldsymbol{\Lambda}_j\boldsymbol{L})\boldsymbol{\xi}_j &= \boldsymbol{\Lambda}_j(\boldsymbol{v}_j - \boldsymbol{K}_+\boldsymbol{s}_j^{\text{in}}). \end{aligned} \tag{102}$$

The first form remains well defined when some gains are zero. Let $\boldsymbol{\Xi} = [\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_V] \in \mathbb{R}^{C\times V}$ and let $\boldsymbol{\xi}_r^\top$ denote its rth row. The value-batched forward substitution is

$$\boldsymbol{\xi}_r = \boldsymbol{\beta}_r \odot \left[\boldsymbol{v}_r - (\boldsymbol{S}^{\text{in}})^\top \boldsymbol{k}_r^+ - \sum_{i<r} L_{ri}\boldsymbol{\xi}_i\right]. \tag{103}$$

The geometry $\boldsymbol{L}$ is shared, but its effective columns are scaled differently for every value coordinate.

To expose the compact-WY factors, define

$$\boldsymbol{A}_j = (\boldsymbol{I}_C + \boldsymbol{L}\boldsymbol{\Lambda}_j)^{-1}, \qquad \boldsymbol{\eta}_j = \boldsymbol{A}_j\boldsymbol{v}_j, \qquad \boldsymbol{W}_j = \boldsymbol{A}_j\boldsymbol{K}_+. \tag{104}$$

The inverse denotes a unit-lower-triangular solve. For the normalized rank-one transition

$$\boldsymbol{E}_{r,j} = \boldsymbol{I}_K - \beta_{r,j}\boldsymbol{g}_r^-(\boldsymbol{k}_r^+)^\top, \tag{105}$$

the compact-WY identity is

$$\boldsymbol{E}_{C,j}\boldsymbol{E}_{C-1,j}\cdots\boldsymbol{E}_{1,j} = \boldsymbol{I}_K - \boldsymbol{G}_-^\top\boldsymbol{\Lambda}_j\boldsymbol{W}_j. \tag{106}$$

The unscaled innovations satisfy

$$\boldsymbol{\varepsilon}_j = \boldsymbol{\eta}_j - \boldsymbol{W}_j\boldsymbol{s}_j^{\text{in}}, \qquad \boldsymbol{\xi}_j = \boldsymbol{\Lambda}_j(\boldsymbol{\eta}_j - \boldsymbol{W}_j\boldsymbol{s}_j^{\text{in}}). \tag{107}$$

The innovations $\varepsilon_{r,j}$ are exactly the exclusive residuals $\delta_{t,j}$ in Equation (27). Hence the predictive likelihood can be evaluated from $\boldsymbol{\varepsilon}$ and the variances $\boldsymbol{\Delta}$ emitted by Stage 1 without another recurrent read.

B.4 Parallel reads, chunk carry, and stable ratios

For a post-write query $\boldsymbol{q}_r$, define

$$[\boldsymbol{Q}_+]_{r,:} = (\boldsymbol{\Gamma}_r^\top \boldsymbol{q}_r)^\top, \qquad \boldsymbol{L}_{qg} = \operatorname{tril}(\boldsymbol{Q}_+ \boldsymbol{G}_-^\top). \tag{108}$$

The diagonal is retained because the current write is visible to the current query. All outputs and the physical outgoing state are

$$\begin{aligned} \boldsymbol{O} &= \boldsymbol{Q}_+ \boldsymbol{S}^{\text{in}} + \boldsymbol{L}_{qg} \boldsymbol{\Xi}, \\ \boldsymbol{S}^{\text{out}} &= \boldsymbol{\Gamma}_C \left(\boldsymbol{S}^{\text{in}} + \boldsymbol{G}_-^\top \boldsymbol{\Xi} \right). \end{aligned} \tag{109}$$

A pre-write read replaces tril by its strictly lower part.

The inverse cumulative decay in $\boldsymbol{g}_r^-$ need not be materialized. Define the physical transition from immediately after token i to token r by

$$\boldsymbol{\Phi}_{r\leftarrow i} = \begin{cases} \boldsymbol{D}_r \boldsymbol{D}_{r-1} \cdots \boldsymbol{D}_{i+1}, & r > i, \\ \boldsymbol{I}_K, & r = i. \end{cases} \tag{110}$$

Then all required interactions have the stable form

$$\begin{aligned} L_{ri} &= \boldsymbol{k}_r^\top \boldsymbol{\Phi}_{r\leftarrow i} \boldsymbol{g}_i, & i < r, \\ (L_{qg})_{ri} &= \boldsymbol{q}_r^\top \boldsymbol{\Phi}_{r\leftarrow i} \boldsymbol{g}_i, & i \le r, \\ \boldsymbol{S}^{\text{out}} &= \boldsymbol{\Gamma}_C \boldsymbol{S}^{\text{in}} + \sum_{i=1}^{C} \boldsymbol{\Phi}_{C\leftarrow i} \boldsymbol{g}_i \boldsymbol{\xi}_i^\top. \end{aligned} \tag{111}$$

For positive diagonal decays, $\boldsymbol{\Phi}_{r\leftarrow i}$ should be formed from differences of cumulative log decays. The decay applied to a write at token i begins at token $i+1$; $\boldsymbol{D}_i$ acts before that write.

Under a time-local gauge transformation

$$\widehat{\boldsymbol{B}}_i \mapsto \lambda_i \widehat{\boldsymbol{B}}_i, \qquad \widehat{\boldsymbol{P}}_i \mapsto \widehat{\boldsymbol{P}}_i / \lambda_i, \tag{112}$$

we have $\boldsymbol{g}_i \mapsto \boldsymbol{g}_i/\lambda_i$ and $\boldsymbol{\beta}_i \mapsto \lambda_i \boldsymbol{\beta}_i$. Hence $L_{ri}\beta_{i,j}$, $\boldsymbol{g}_i \xi_{i,j}$, and all physical outputs in Equations (102)–(111) are gauge invariant.

B.5 Two-stage schedule, cost, and exactness

The complete forward schedule is:

1. **Coupled-factor stage.** Traverse tokens causally with Equations (82)–(83), and evaluate the projection either by alternating Equation (84) or by the scalar form in Equations (88)–(93). Emit $\boldsymbol{g}_t$ and $\boldsymbol{\beta}_t$, optionally retain $\boldsymbol{\Delta}_t$ for the predictive likelihood, and carry the projected factors.
2. **Conditional-mean stage.** For each memory chunk, form the two-key interactions in Equation (99), solve Equation (102) batched over value coordinates, evaluate Equation (109), and carry $\boldsymbol{S}^{\text{out}}$ to the next chunk.

With N_{proj} reverse-KL sweeps per token, Stage 1 costs

$$O(T(N_{\text{proj}} + 1)(K + V)) \tag{113}$$

work and has sequential depth proportional to $T(N_{\mathrm{proj}}+1)$. Storing the factored gain sequence costs $O(T(K+V))$; checkpointing may instead recompute it during backpropagation. The scalar evaluation has the same asymptotic reduction work, with N_{proj} scalar sweep steps but only $N_{\mathrm{proj}}-1$ required key-ratio updates, and reconstructs the factor vectors once per token. For a converged solve, replace N_{proj} by the number N_{root} of safeguarded scalar root iterations. In either case the token-to-token depth remains linear in T.

For one Stage 2 chunk, constructing the shared interactions costs $O(C^2K)$, the value-batched triangular systems cost $O(C^2V)$, and multiplication by the carried memory costs $O(CKV)$. Across the sequence, the direct implementation costs

$$O(TC(K+V)+TKV) \tag{114}$$

with $O(KV+C^2+CV)$ working storage beyond inputs and outputs. In the direct schedule, each chunk's C-row solve waits for $\boldsymbol{S}^{\mathrm{in}}$, so the per-trajectory critical path remains $\Theta(T)$, vectorized over values. As in Appendix A, precomputing distinct $\boldsymbol{W}_j$ factors moves those local solves off the critical path and leaves T/C chunk carries, but costs $O(VC^2K)$ work and $O(VCK)$ storage per chunk. If $\beta_{r,j}=\beta_r$ for every j, Stage 2 reduces to one shared DeltaNet/KDA triangular system with V right-hand sides. Tying gains within G value groups requires G systems and is exact only for the corresponding grouped uncertainty model.

Exactness. Assume positive diagonal decays and let Stage 1 use the same prediction, gauge, and reverse-KL solution as the sequential KDN-2 reference. Covariance updates are independent of the realized observation, so Stage 1 may run ahead of the mean recurrence and emits the same $\boldsymbol{g}_t$ and $\boldsymbol{\beta}_t$ by causal induction. For fixed gains, Equation (98) gives $\boldsymbol{y}_{r-1,j}=\boldsymbol{s}_j^{\mathrm{in}}+\sum_{i<r}\boldsymbol{g}_i^{-}\xi_{i,j}$. Substitution into $\varepsilon_{r,j}$ yields the unit-lower system in Equation (101), whose unique solution equals the sequential innovations. Substituting back gives the same prefix and outgoing states; applying the queries gives Equation (109). Induction over chunk boundaries therefore proves equality of posterior means, predictive residuals, outputs, and boundary states.